

Companion ledger: definitions, lemmas and proofs

Team Ninja Turtles

Theory: Householder complexity of one-layer state tracking

Status legend: **[Lean]** machine-checked (expressivity/lean), **[hand]** proved by hand in this file, not machine-checked (evidence level hypothesis in the paper until reviewed), **[cert]** every concrete instance re-verified by the exact verifier.

Setting

G finite group, $\Sigma \subseteq G \setminus \{e\}$ generating, word $w = s_1 \dots s_T$, target $g_t = s_t \dots s_1$. A *one-layer realisation* consists of a state space $V = \mathbb{R}^{d \times m}$, maps $A: \Sigma \rightarrow \mathbb{R}^{d \times d}$, $B: \Sigma \rightarrow V$, an initial state h_0 and a readout $\phi: V \rightarrow G$ with $h_t = A(s_t) h_{t-1} + B(s_t)$ and $\phi(h_t) = g_t$ for all words and all t . It is *finite-state* if the set Q of reachable states is finite, and it is *k-Householder* if $\text{rank}(A(s) - I) \leq k$ for every letter (this contains DeltaProduct with k factors $I - \beta u u^T$, $\beta \in [0, 2]$, by L1).

Definitions. $h(G, \Sigma) = \min \text{ over faithful real representations } \rho \text{ of } G \text{ of } \max_s \text{rank}(\rho(s) - I)$. $h(G, \Sigma) = \min \text{ over } (H, \pi, (t_s), \rho) \text{ of } \max_s \text{rank}(\rho(t_s) - I)$, where H is a finite group, $\pi: H \rightarrow G$ a surjective homomorphism, $t_s \in H$ with $\pi(t_s) = s$ generating H , and ρ a faithful real representation of H . Clearly $h \leq h$ (take $H = G$).

L1 (rank lemma) [Lean]

If $R_1, \dots, R_k$ have $\text{rank} \leq 1$ then $\text{rank}(\prod_i (I + R_i) - I) \leq k$. In particular a product of k generalized Householder matrices differs from I by $\text{rank} \leq k$. Lean: `rank_prod_sub_one_le`, `deltaproduct_rank_le`; `tightness: strengthened_bound_false`.

L2 (compression lemma; necessity) [hand]

Claim. If a finite-state one-layer realisation of (G, Σ) has $\text{rank}(A(s) - I) \leq k$ for all letters, then $h^*(G, \Sigma) \leq k$.

Proof. For a word w let α_w be the affine map $h \rightarrow A(w)h + B(w)$ obtained by composing the letter maps ($A(w) = A(s_T) \dots A(s_1)$). Q is invariant under every α_s , so the α_w restricted to Q form a finite transformation monoid T , generated by the restrictions of the α_s . Define $\psi: T \rightarrow G$ by $\psi(\alpha_w|_Q) = g(w)$. It is well defined: if $\alpha_w|_Q = \alpha_{w'}$ then $g(w) = \phi(\alpha_w(h_0)) = \phi(\alpha_{w'}(h_0)) = g(w')$ ($h_0 \in Q$). It is a monoid homomorphism onto G (concatenation; Σ generates G). Let K be the minimal two-sided ideal of the finite monoid T and $e \in K$ an idempotent (exists in every finite semigroup). Then $\Gamma = eTe$ is a group with identity e (a maximal subgroup of the completely simple semigroup K). $\psi(e)$ is an idempotent of G , hence $\psi(e) = 1$, and $\psi(e t e) = \psi(t)$, so ψ restricted to Γ is onto G . Fix a word u with $\alpha_u|_Q = e$. Let $Q_e = e(Q) = \alpha_u(Q)$, W its affine span and U the direction space of W . e fixes Q_e pointwise ($e(e(q)) = e(q)$), so α_u fixes W pointwise and $A(u)$ acts as the identity on U . For $s \in \Sigma$ put $x_s = e \alpha_s e \in \Gamma$; $\psi(x_s) = \psi(\alpha_s) = s$, so the subgroup Γ' generated by the x_s maps onto G . Every element of Γ maps Q_e bijectively to itself (Γ is a group with identity e , and e is the identity on Q_e), hence its affine extension $\alpha_{\{u v u\}}$ maps W to W ; the restriction to W depends only on the element of Γ (affine maps agreeing on the affinely spanning set Q_e agree on W), so we get a homomorphism $\Gamma \rightarrow \text{Aff}(W)$. It is injective: an element acting trivially on W acts trivially on Q_e , and an element $t = ete$ of Γ is determined by its action on Q_e ($t(q) = t(e(q))$ for all $q \in Q$). Taking linear parts gives a homomorphism $\lambda: \Gamma \rightarrow \text{GL}(U)$. It is injective as well: if $\lambda(t) = I$ then t acts on W as a translation; a translation of finite order is the identity. Rank bound: the linear part of x_s on U is $\lambda(x_s) = A(u) A(s) A(u)|_U = A(u) A(s)|_U$ ($A(u) = I$ on U), and for $y \in U$, $\lambda(x_s) y - y = A(u) A(s) y - A(u) y = A(u) (A(s) - I) y$, so $\text{rank}(\lambda(x_s) - I_U) \leq \text{rank}(A(s) - I) \leq k$. Matrix-valued states: $V = \mathbb{R}^{d \times m}$ with A acting from the left. Let $C \subseteq \mathbb{R}^d$ be the span of all columns of all elements of U . C is invariant under $A(u)$ and under $A(u)A(s)$ for elements of the group (columns of $A H$ are A times columns of H); a word acts trivially on U iff it acts trivially on C . Hence $\lambda'(t)$

= (linear part)| C is a faithful representation of Γ on C , and $\text{rank}(\lambda'(x_s) - I_C) \leq \text{rank}(A(u)(A(s) - I)) \leq k$. (For vector states $m = 1$ and $C = U$.) Finally take $H = \Gamma'$, $\pi = \text{psl}\{\Gamma'\}$, $t_s = x_s$, $\rho = \lambda'$ restricted to Γ' ; ρ is faithful, $\pi(t_s) = s$, the t_s generate H , and $\text{rank}(\rho(t_s) - I) \leq k$. Hence $h(G, \Sigma) \leq k$. *QED.* Remarks.* (i) No invertibility of $A(s)$ is needed ($\beta = 1$, projections, are allowed). (ii) The additive input term $B(s)$ and the readout are arbitrary. (iii) The finite-state assumption is essential: without a bound on the reachable set, infinite matrix groups (free groups generated by two rotations) give other behaviour (analogue_candidates.md); finite precision makes every implemented recurrence finite-state but not exactly linear.

L3 (sufficiency) [hand, cert]

If $(H, \pi, (t_s), \rho)$ has $\text{rank}(\rho(t_s) - I) \leq k$, a finite-state k -Householder realisation exists in dimension $\dim \rho$ with $\beta \in \{0, 2\}$: choose an H -invariant inner product (average any inner product over H) and an orthonormal basis for it, so ρ is orthogonal; by Cartan–Dieudonné (Scherk’s form) each $\rho(t_s)$ is a product of exactly $\text{rank}(\rho(t_s) - I)$ reflections; pad with identity factors ($\beta = 0$). Take h_0 outside the finite union of the proper subspaces $\text{Fix}(\rho(x))$, $x \neq 1$ (ρ faithful), so $x \rightarrow \rho(x) h_0$ is injective on H ; the readout maps $\rho(x) h_0$ to $\pi(x)$. Every concrete instance used in this work is re-verified by `realisation.verify` (exact closure, consistent readout, invariant form, rank, explicit reflection factorisation).

Theorem 1. For every (G, Σ) : a finite-state one-layer k -Householder realisation exists iff $k \geq h^*(G, \Sigma)$. (L2 + L3.)

L4 (order lemma) [hand]

If some letter s has order ≥ 3 then $h(G, \Sigma) \geq 2$. Proof.* Suppose $\text{rank}(\rho(t_s) - I) \leq 1$ with ρ faithful on the finite group H . If the rank is 0 then $t_s = 1$ and $s = \pi(1) = 1$, impossible. Otherwise $M = \rho(t_s) = I + R$ with $\text{rank } R = 1$, $R = v w^T$, $R^2 = \tau R$ with $\tau = w^T v$, and $M^n = I + c_n R$ with $c_n = ((1 + \tau)^n - 1)/\tau$ ($c_n = n$ if $\tau = 0$). M has finite order N , so $c_N R = 0$, hence $c_N = 0$: $\tau \neq 0$ and $(1 + \tau)^N = 1$ with $1 + \tau$ real, so $1 + \tau = -1$ ($1 + \tau = 1$ means $\tau = 0$). Then $M^2 = I + c_2 R = I + (((-1)^2 - 1)/(-2)) R = I$, i.e. $t_s^2 = 1$ and $s^2 = \pi(t_s)^2 = 1$, contradicting $\text{ord}(s) \geq 3$. *QED.*

L5 (diagonal families) [hand]

In L2 the compressed maps are $A(u) A(s)$ restricted to an invariant subspace. If all $A(s)$ are diagonal (real or complex), the compressed maps commute, so Γ' is abelian and $G = \pi(\Gamma')$ is abelian. If the diagonal entries are real, each compressed map is a restriction of a real diagonal matrix of finite order on its invariant subspace, so its eigenvalues are real roots of unity (± 1) and it is diagonalisable: Γ' is an elementary abelian 2-group, hence so is G . If the entries lie in $[0, 1]$, the only real root of unity available is 1, so Γ' is trivial and $G = 1$. Conversely an abelian G is realised by the count cover (one coordinate per letter, complex rotation by $2\pi / \text{ord}(s)$), and an elementary abelian 2-group by real ± 1 coordinates. *Statement:* one-layer finite-state realisations exist for diag_pos iff $G = 1$, for diag_pm iff G is an elementary abelian 2-group, for cdiag iff G is abelian. (Consistent with Grazzi et al. 2024, Sarrof et al. 2024, Shakerinava et al. 2026; scout_evidence.md.)

L6 (computability of h) [standard]

$\text{codim Fix } \rho(g) = \dim \rho - (1/|g|) \sum_j \chi_\rho(g^j)$; codimensions add over direct sums; a sum of real irreducibles is faithful iff the intersection of their kernels is trivial. Hence $h(G, \Sigma) = \min$ over sets S of real irreducibles with trivial common kernel of $\max_s \sum_{\psi \in S} \text{codim}_\psi(s)$: a finite optimisation over the real character table (chartab.py, GAP).

L7 (abelian groups) [hand, cert]

For abelian G : $h(G, \Sigma) = 1$ if every letter is an involution, else 2. Proof.* Lower bound: $G \neq 1$ and L4. Upper bound (count cover): $H = \prod_{s \in \Sigma} Z_{\text{ord } s}$, $t_s =$ generator of the s -th factor, $\pi(t_s) = s$ (a homomorphism because G is abelian), $\rho =$ sum over letters of a reflection ($\text{ord } s = 2$) or a planar rotation by $2\pi / \text{ord}(s)$ (rank 2). Certified instances: Z_3/all ($|H| = 9$), Z_2^3/all ($|H| = 128$).

Corollaries (each instance certified by the verifier)

- **C1 (A5, the canonical NC1-complete word problem):** $h(A_5, \text{all}) = 2$ (icosahedral rotation group, dimension 3) and $h(A_5, \text{involutions}) = 1$ although $h(A_5, \text{involutions}) = 2$: the full icosahedral reflection group $H_3 = A_5 \times Z_2$ ($|H| = 120$) maps onto A_5 with every involution lifted to a reflection (minus a rotation by π). Plain DeltaNet (one reflection per token) tracks A_5 exactly.

- **C2 (alphabet dependence):** $h(S5, \text{transpositions}) = 1$ but $h(S5, \text{all}) = 4$ and $h(S5, \text{all}) \in [2, 4]$; the same group, different inputs.
- **C3 (non-monotonicity in circuit complexity):** the solvable groups $Q8$ and $SL(2,3)$ have $h = 4$ (bounds $[2, 4]$), while the non-solvable $A5$ has $h^* = 2$; $Z2^k$ (in $AC0[2]$) has $h = k$ but $h^* = 1$.
- **C4 (representation matters):** $h^*(S4, \text{all}) = 2$ (cube rotation group) whereas the permutation representation needs 3 (4-cycles).

Open

$h(S5, \text{all})$, $h(Q8, \text{all})$, $h^*(SL(2,3), \text{all})$ (gaps between the lemma lower bound 2 and the best certified upper bound); multi-layer trade-offs between depth and Householders; the effect of the short convolution; learnability.